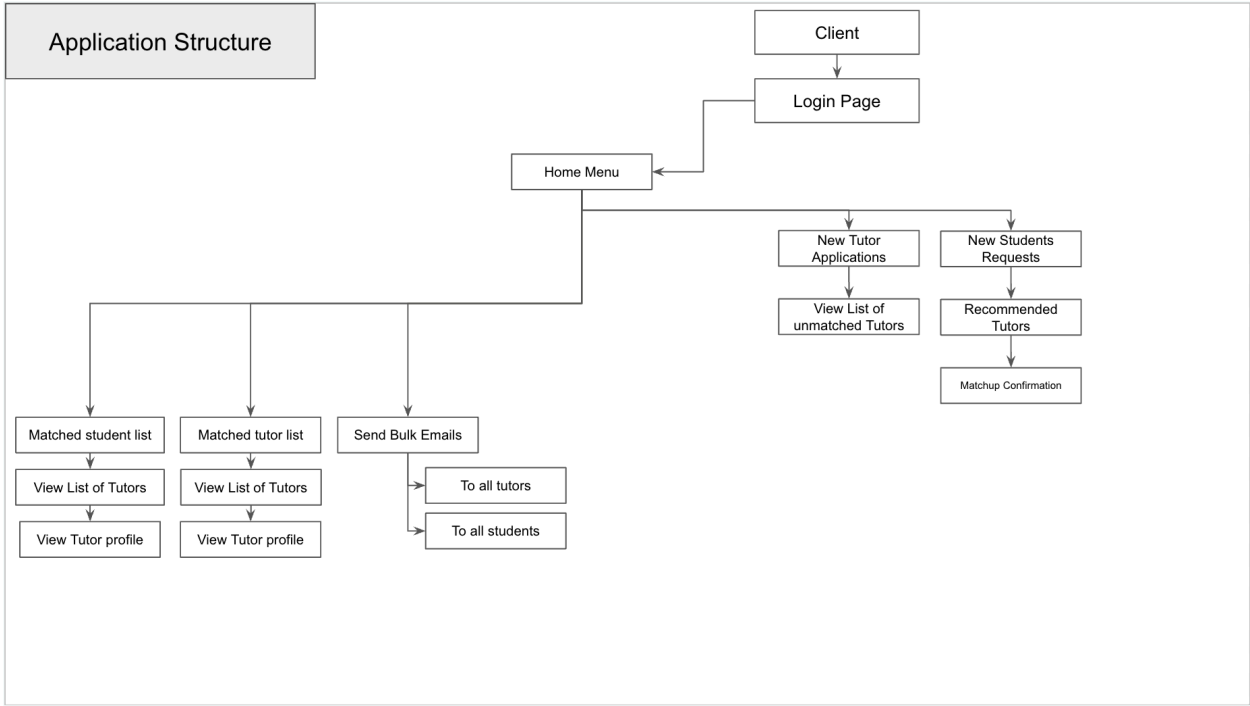
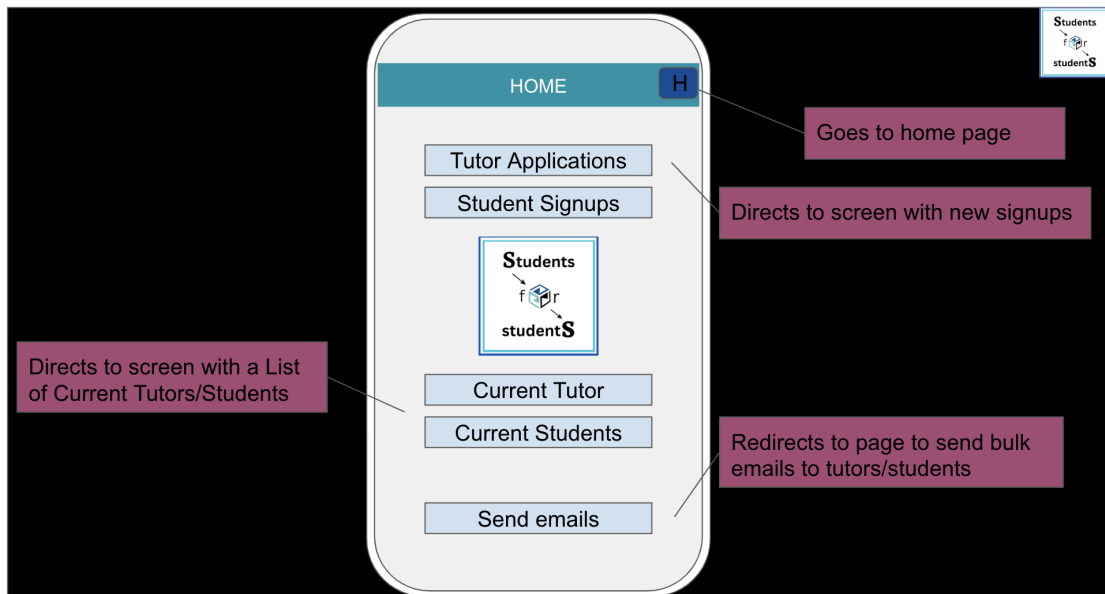
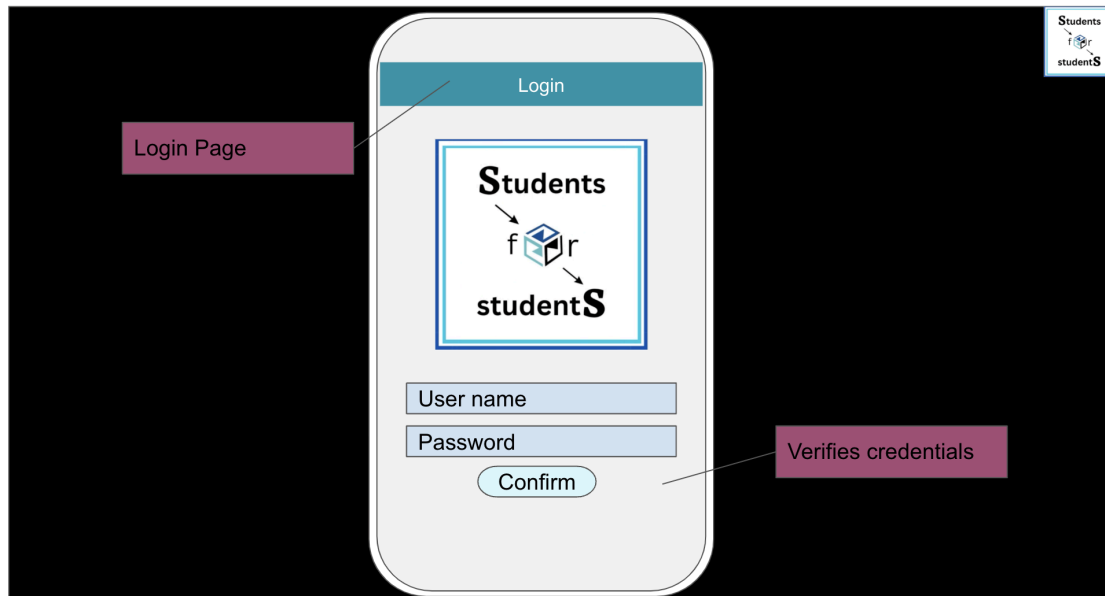


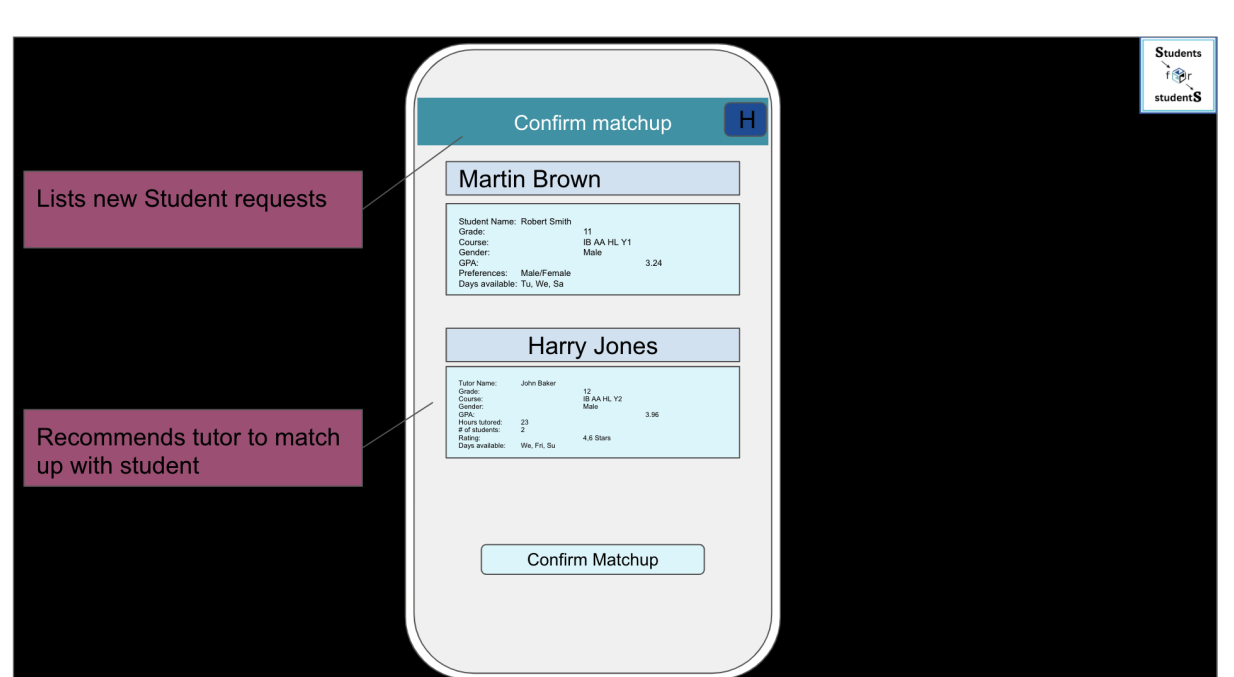
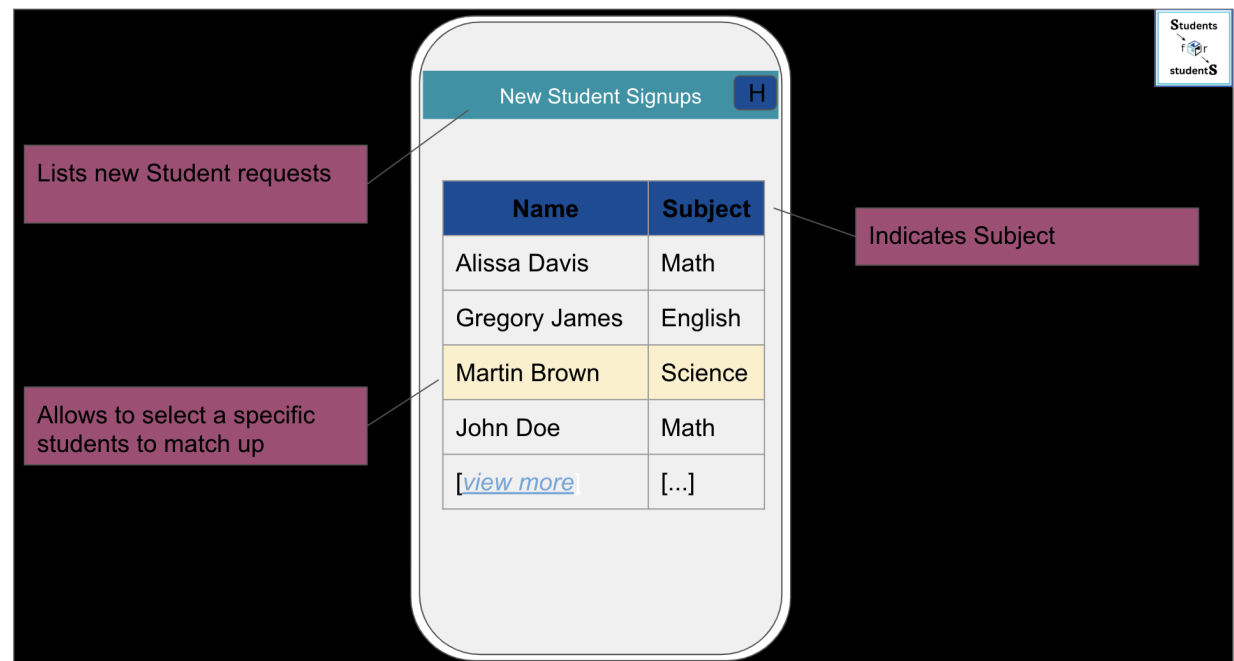
Design

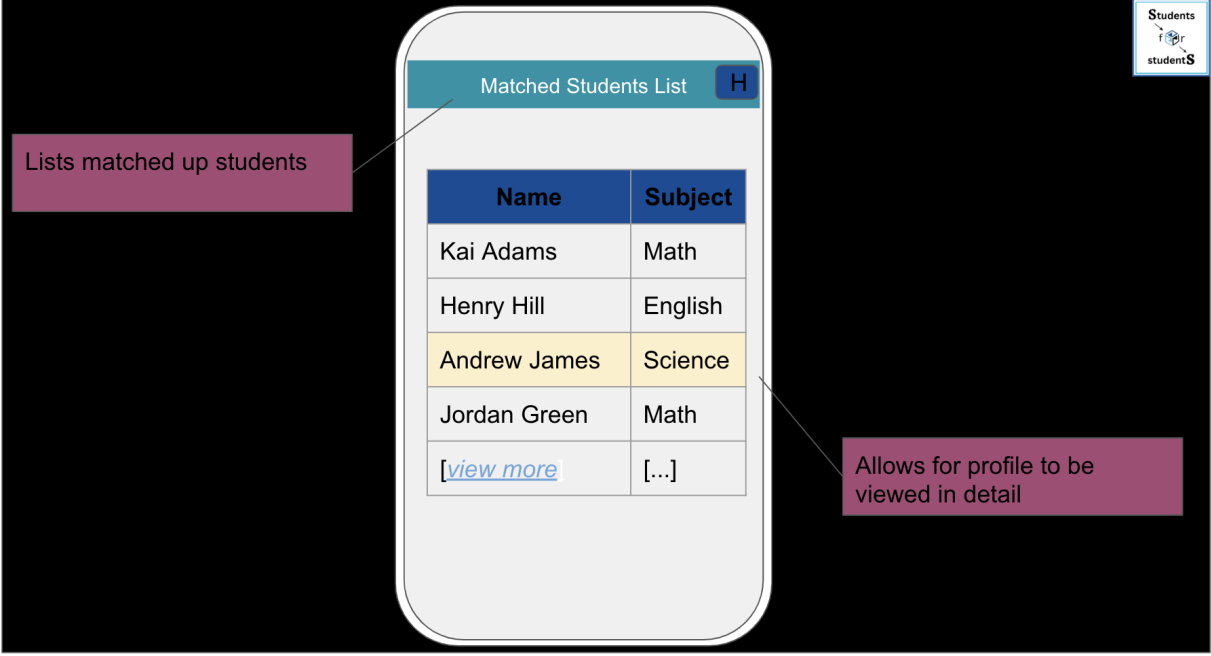
Structure Chart



Wireframes







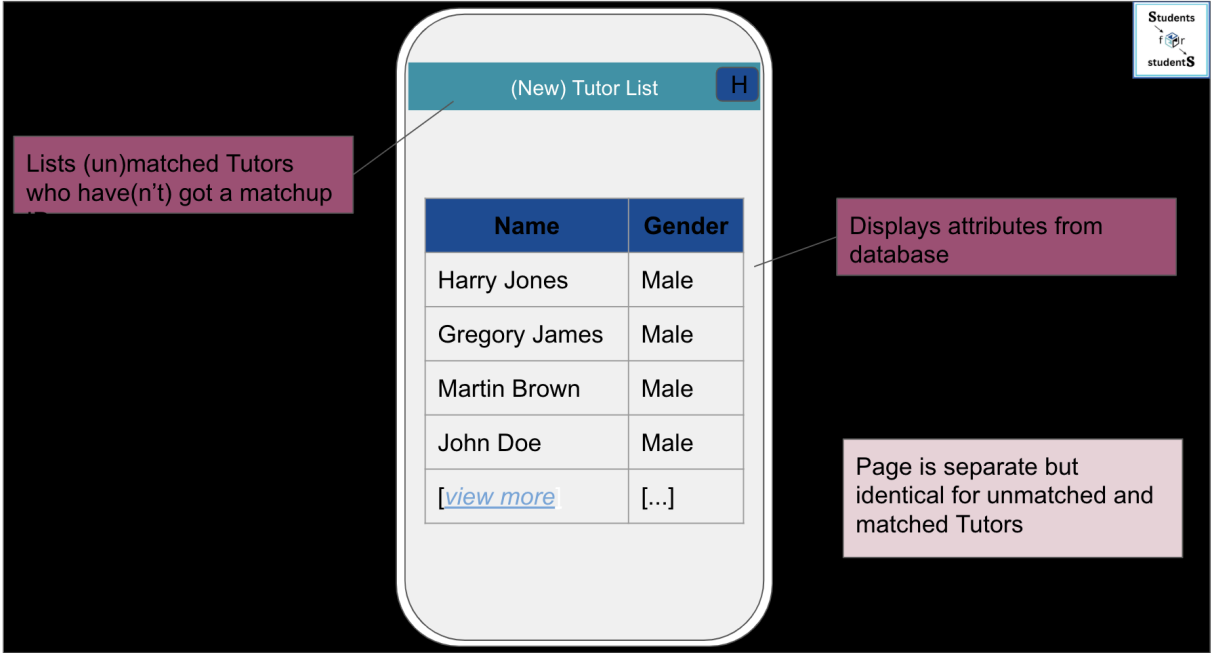
Students for studentS

Matched Students List H

Lists matched up students

Name	Subject
Kai Adams	Math
Henry Hill	English
Andrew James	Science
Jordan Green	Math
[view more]	[...]

Allows for profile to be viewed in detail



Students for studentS

(New) Tutor List H

Lists (un)matched Tutors who have(n't) got a matchup

Name	Gender
Harry Jones	Male
Gregory James	Male
Martin Brown	Male
John Doe	Male
[view more]	[...]

Displays attributes from database

Page is separate but identical for unmatched and matched Tutors

Detailed profile of a student/tutor

Profile

H

Andrew James

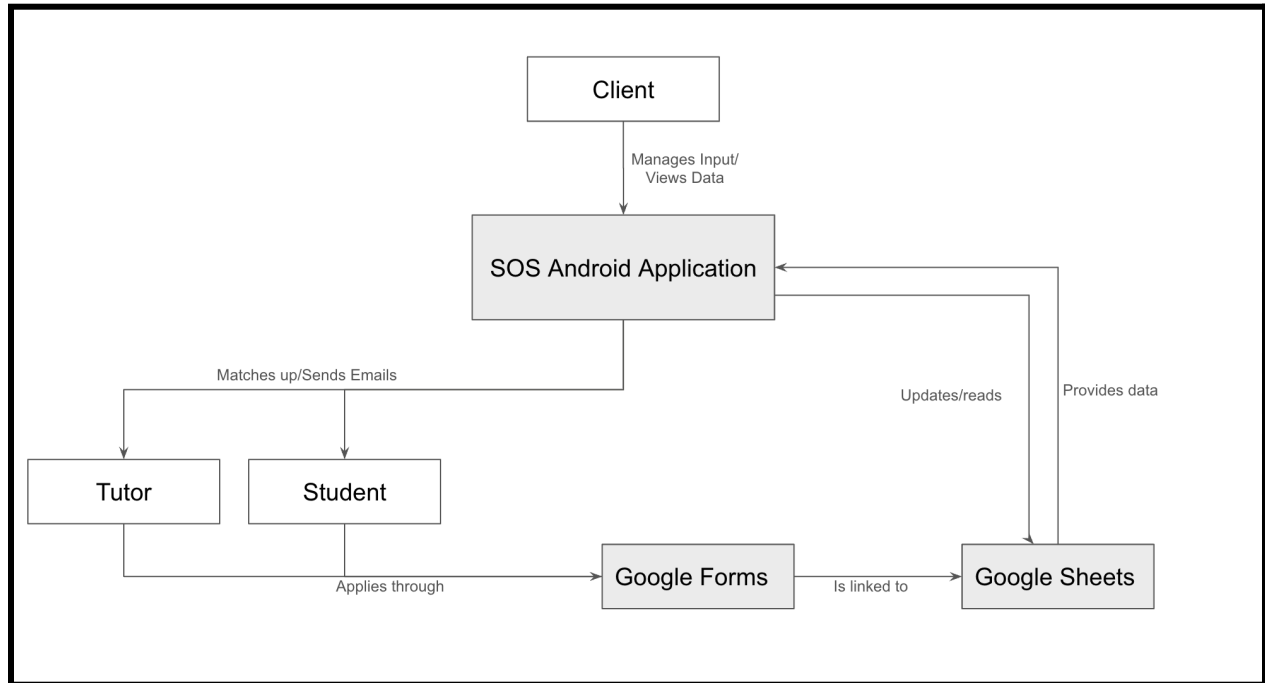
Subject	Math
ID	12345
Matchup ID	TA-34998
Gender	Male
Grade	Senior
Email	andrew.j@gmail.com

Students

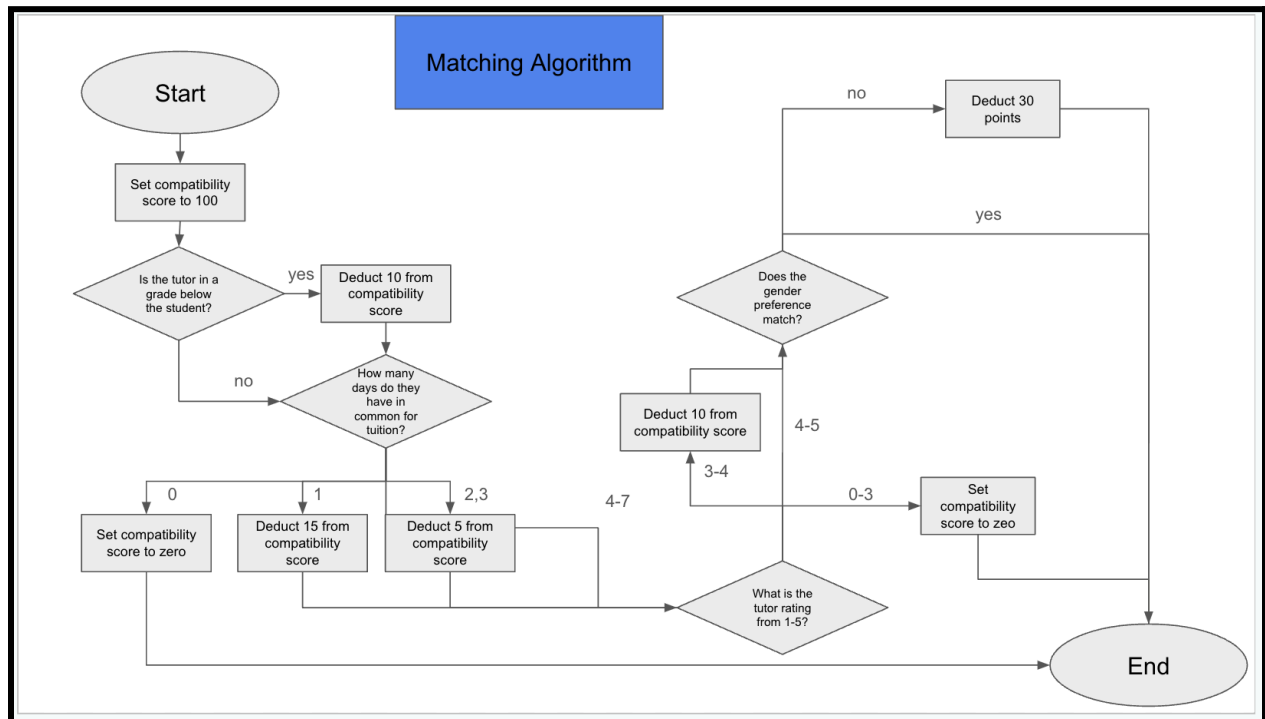
For

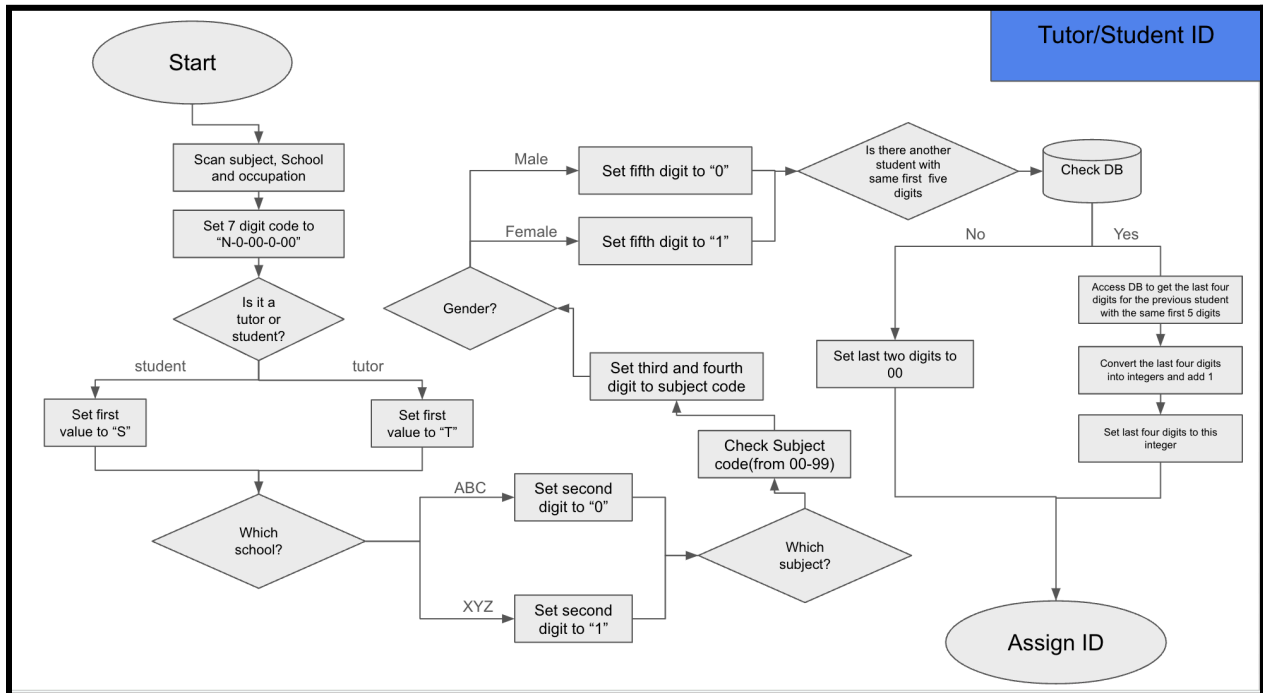
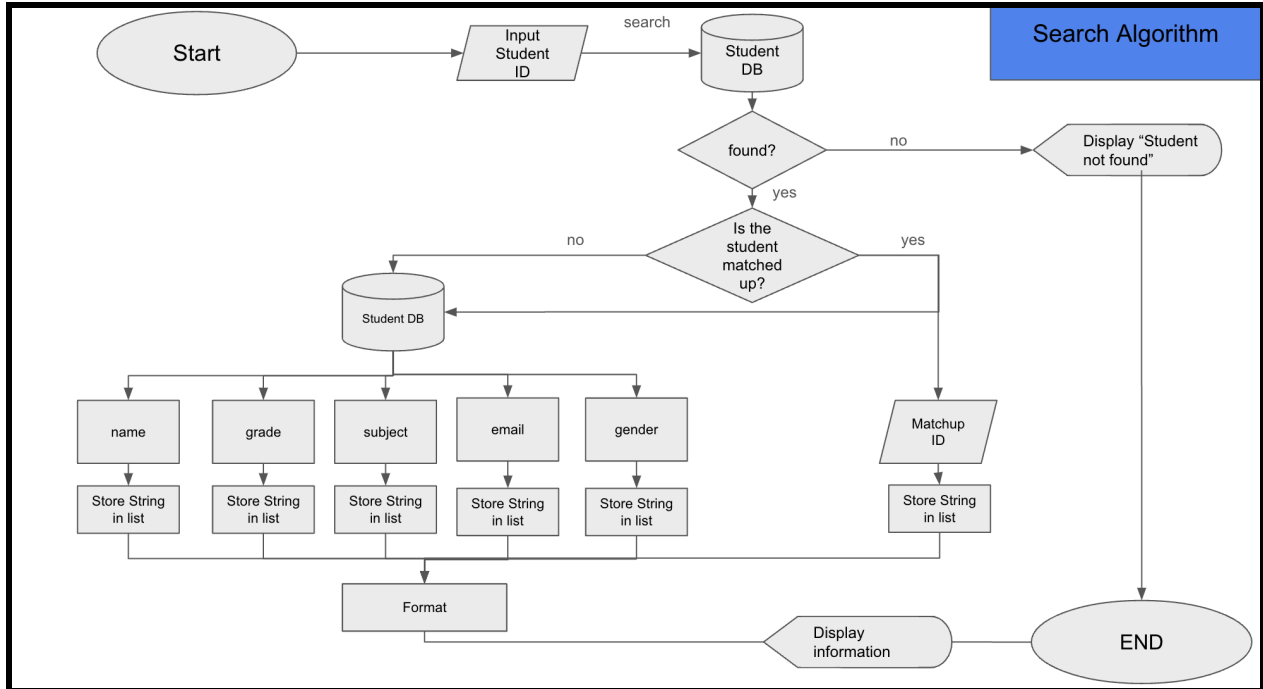
studentS

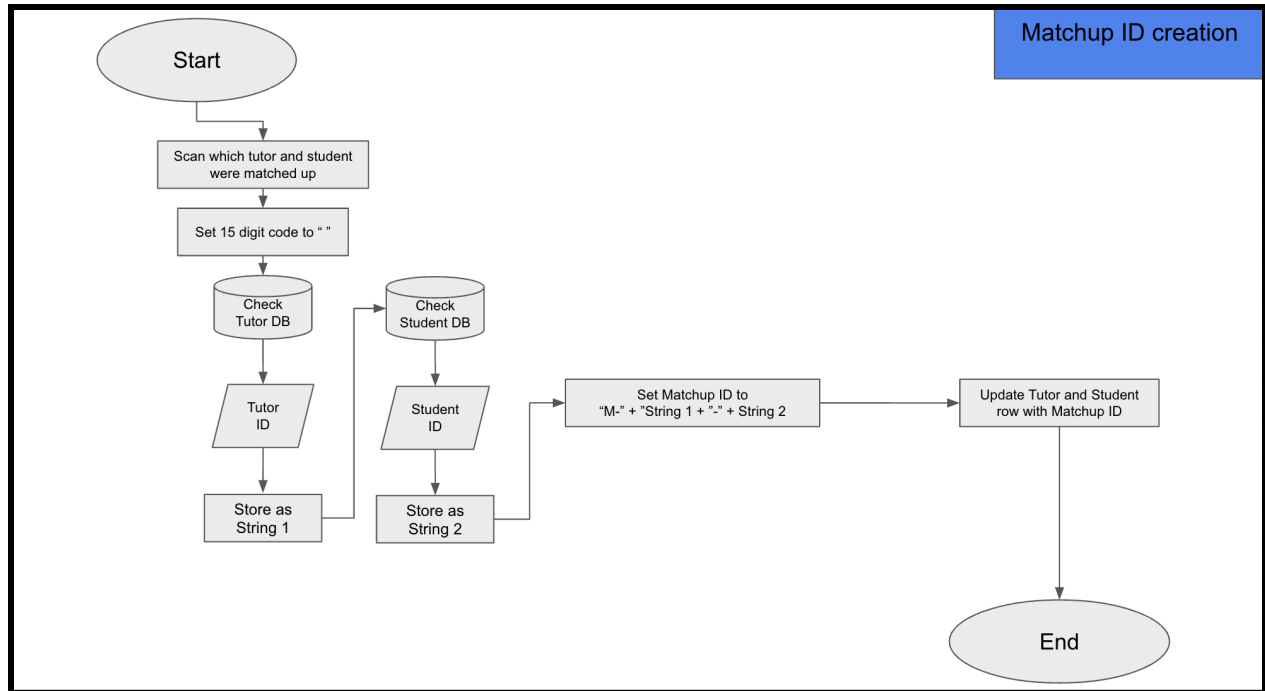
DFD Level 0 Context Diagram



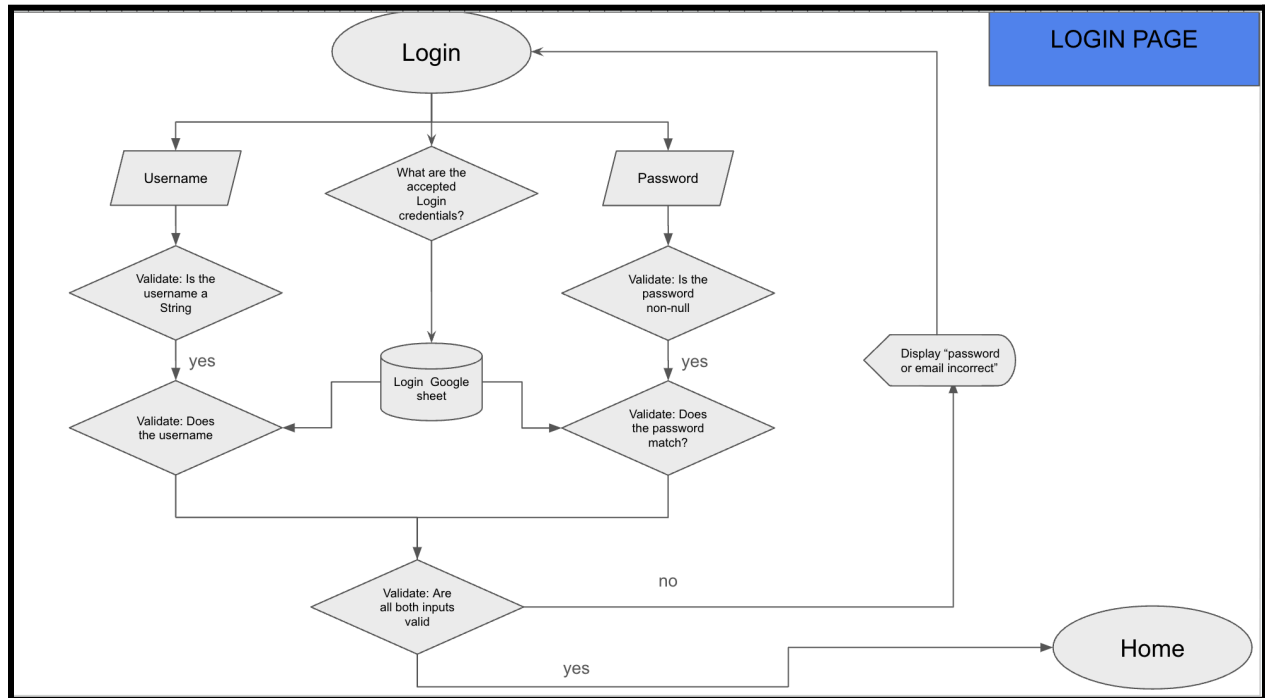
CORE Algorithms







System Flow chart



UML Diagram

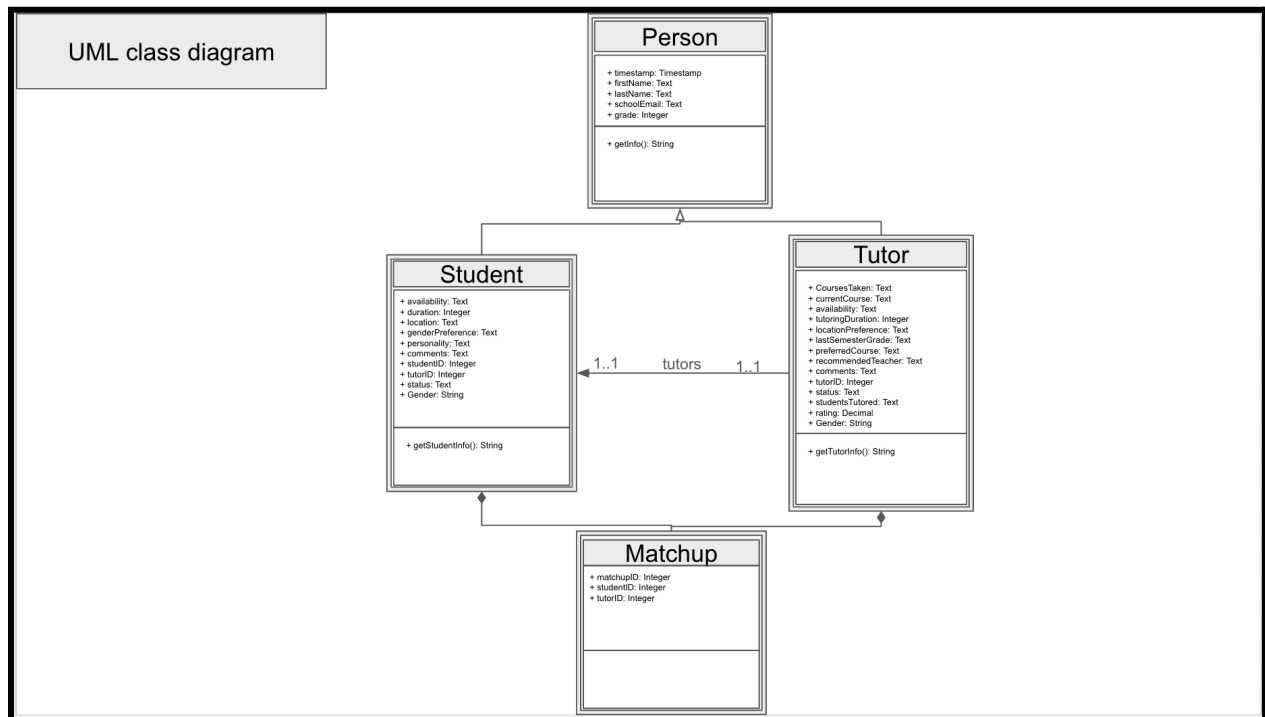


Table structures

Table 1: Subject-specific high school student information

Field Name	Data Type	Validation Rules	Description
Timestamp	Timestamp	Non-null	The time at which the request was submitted
Email Address	Text	Unique, valid email format	The email address of the student requesting tutoring
First Name	Text	Non-empty	The student's first name
Last Name	Text	Non-empty	The student's last name
School Email	Text	Unique, valid email format	The student's school email address
Grade	Grade Level	Positive integer	The student's grade level
Course	Text	Non-empty	The specific course the student needs help with
Last Semester Grade	Dropdown text	One of: "A", "B", "C", "D", "E", "F", " does not apply"	The student's grade in the course last semester (if applicable)
Availability	Text	Options: "Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"	A list of days and times the student is available for tutoring
Duration	Integer	Positive integer	The desired duration of the tutoring sessions
Location	Text	One of: "online", "in person"	Whether the student prefers online or in-person tutoring
Gender Preference	Text	One of: "male", "female"	The student's gender preference for the tutor
Personality	Text	Optional	The student's personality type (e.g., introvert, extrovert, energetic)
Comments	Text	Optional	Any additional comments from the student
Gender	Text	One of: "male", "female"	The student's gender
Subject	Text	Non-empty	The subject in which help is required
Non-input data			
Student ID	Integer	Unique Assigned ID	The ID of the student for the specific subject

Matchup ID	Integer	Unique ID for Student and Tutor	The ID of the tutor-student matchup
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Table 2: Subject-specific tutor information

Field Name	Data Type	Validation Rules	Description
Timestamp	Timestamp	Non-null	The time at which the request was submitted
Email Address	Text	Unique, valid email format	The email address of the tutor requesting to tutor
First Name	Text	Non-empty	The tutor's first name
Last Name	Text	Non-empty	The tutor's last name
History Courses Taken	Text	Non-empty	A list of history courses the tutor has taken
Current Course	Text	Non-empty	The history course the tutor is currently taking
Availability	Checkbox	Non-empty	A list of days and times the tutor is available for tutoring
Tutoring Duration	Integer	Positive integer	The maximum duration the tutor is willing to tutor a student
Location Preference	Text	One of: "online", "in person"	Whether the tutor prefers online or in-person tutoring
Last Semester Grade	Text	One of: "A", "B", "C", "D", "E", "F", "does not apply"	The tutor's grade in the current history course
Preferred Course	Text	Non-empty	In the history course, the tutor feels most comfortable tutoring students from
Recommended Teacher	Text	Optional	The name of a teacher who can represent the tutor's strengths and personality
Counselor	Text	Optional	The name of the student's high school counselor
Gender	Text	One of: "male", "female"	The student's gender
Comments	Text	Optional	Any additional comments from the tutor

Non-input data			
Tutor ID	Integer	Unique identifier	The unique ID assigned to the tutor for each subject
Matchup ID	Integer	Unique ID for Student and Tutor	The ID of the tutor-student matchup
Rating	Float	Between 0-5	Feedback that is given by students

Table 3: Login credentials

Field Name	Data Type	Validation Rules	Description
Email	Text	Non-empty	The email address of the admin
Passcode	Text	Non-empty	The unique passcode provided by the client

Test plan

Success criteria	Test Case	Actions	Expected Result
Is able to authenticate users through login credentials	1.1	Correct Email and Password	The new tutor should be added successfully and appear in the list of available tutors.
	1.2	Invalid Password	Error notification
	1.3	Invalid Email	Error notification
Matches students correctly with suitable tutors using a compatibility algorithm with at least a 90% success rate	2.1	Let the algorithms rank the potential matchups and then let the client review and rank the matchups.	At least 90% of the time the algorithm should align with what the client would have chosen.

Reduces the time my client takes to match students with tutors to under 1,5 min	3.1	Measure the average time taken from logging into the app and updating a new matchup out of 5 trials.	The tutor's preferences should be updated successfully in under 1.5 min
	3.2	Measure the fastest possible time a client could complete this process.	The system should complete the matching process in less than a minute.
Imports student/tutor/login data from Google Sheets and changes are registered in real-time	4.1	Check the time at which changes in the Google Sheets are registered in the App	Within 5 seconds changes in Google sheet data should be read into the Application
Updates Google Sheets with matchup ID once students get matched up	5.1	Check the tutor database after a matchup	The created Matchup ID should be displayed in the tutor's column for Matchup ID
	5.2	Check the student database after a matchup	The created Matchup ID should be displayed in the student's column for Matchup ID
Displays a list of all current matched tutors/students, unmatched tutors	6.1	Open List from Home Page	All tutors/students with a matchup ID as well as unmatched tutors are listed based on a few identifying attributes
Displays a list of all current unmatched students that can be matched	7.1	Open the New Students Page from the Home Page	All students without a matchup ID are listed based on a few identifying attributes. One of the students can be selected and then matched up.
Has an easily navigable graphical user interface (based on client feedback)	8.1	Give it to a "Students for Students" admin without instructions and observe the extent to which they can navigate the features.	The admin should be able to navigate the app after a few minutes without much prior instruction.
	8.2	Get direct feedback from Madalena about its usability.	Madalena provides positive feedback on usability.
Includes error handling to manage invalid data entry from the client	9.1	Appropriate error messages for faulty input	The app displays appropriate error messages for faulty input.
	9.2	Error handling for faulty imported data	app handles faulty imported data and produces an error message, or prevents the data from being imported.

Can send emails to all tutors or students	10.1	Check if all tutors are emailed	Valid tutor addresses are stored as the recipients of the message and duplicate email addresses are combined as one.
	10.2	Check if all students are emailed	Valid student addresses are stored as the recipients of the message and duplicate email addresses are combined as one.